Project Diary – Weekly Progress Overview

Deep Learning for AI-Generated Image Detection: A Review of CNN, Vision Transformer, and Explainable Approaches

Group	148	Course	NTCC Minor Project ETMN100
Programme	B.Tech, Computer Science and Engineering, Semester 7	Project window	20 July – 9 October 2026 (12 weeks)
Institution	Amity School of Engineering and Technology, Amity University Uttar Pradesh	Faculty guide	Dr. Richa Gupta
Team	Rohit Yadav (A2305223472), Vishal (A2305223568), Hardik Mehlawat (A2305223449)		

Weekly overview compiled from our Weekly Progress Reports (WPRs 1–12) and the NTCC project diary.

Week	Dates (2026)	Planned work	Progress / what was done	Output
1	20–26 Jul	Fix the title and scope. Write the search protocol with inclusion and exclusion criteria. Set the section outline.	Scope fixed as real versus AI-generated still images. Search protocol, criteria and an eight-section outline written; working repository set up. Title finalised on 29 Jul (reported in WPR 1).	Search protocol; outline; WPR 1 (dated 29 Jul)
2	27 Jul – 2 Aug	First round of literature collection. Screening by title and abstract.	Eight query seeds run on arXiv, publisher and conference pages and Google Scholar. 101 records found, 26 duplicates removed, 75 screened by title and abstract; 48 primary studies shortlisted.	Candidate and screening lists with stage counts; WPR 2 (dated 5 Aug)
3	3–9 Aug	Read the primary studies in full. Fill the extraction table.	All 48 primary studies read in full and entered in an eight-column table (10 doubtful cells flagged for checking). One preprint with 1 citation excluded; 74 papers kept (48 primary, 11 secondary, 15 background).	Extraction table (48 studies); updated screening list; WPR 3 (dated 12 Aug)
4	10–16 Aug	Finalise the taxonomy. Draft the introduction.	Taxonomy fixed: 5 families (14 / 19 / 2 / 8 / 4); three studies fit two families, and the four dataset studies are kept separate. Introduction drafted. The 10 flagged cells settled from the full texts. *Done 10–12 Aug (NTCC diary), inside this week; reported in this week's WPR.*	Taxonomy map; introduction draft; WPR 4 (dated 19 Aug)
5	17–23 Aug	Draft the background on GANs, diffusion models and the artefacts they leave.	Background section drafted. BibTeX reference file built for all 74 included papers; every identifier opened and checked (74 of 74); one wrong publication year corrected. *Done ahead of schedule on 10–12 Aug (NTCC diary); reported in this week's WPR.*	Background draft; BibTeX file; verification log; WPR 5 (dated 26 Aug)
6	24–30 Aug	Draft the handcrafted and frequency-domain section and the CNN and transfer-learning section (3.1–3.2).	Sections 3.1 and 3.2 drafted, covering the 14 and 19 studies assigned to them. *Done ahead of schedule on 13–14 Aug (NTCC diary); reported in this week's WPR.*	First two taxonomy sections; WPR 6 (dated 2 Sep)
7	31 Aug – 6 Sep	Draft the transformer, multimodal and explainability sections (3.3–3.5).	Sections 3.3 to 3.5 drafted, covering 2, 8 and 4 studies; the taxonomy section is now complete. *Done ahead of schedule on 13–14 Aug (NTCC diary); reported in this week's WPR.*	Complete taxonomy draft; WPR 7 (dated 9 Sep)
8	7–13 Sep	Draft the datasets and evaluation-metrics sections, with the dataset table and figures.	Datasets and metrics sections drafted. Dataset table (11 datasets), eight metrics written as equations, and two figures (search flow, metrics overview) made. *Done ahead of schedule on 13–14 Aug (NTCC diary); reported in this week's WPR.*	Datasets and metrics drafts; dataset table; two figures; WPR 8 (dated 16 Sep)

Week	Dates (2026)	Planned work	Progress / what was done	Output
9	14–20 Sep	Build the master comparison table. Draft the comparative analysis.	Master table of all 48 studies built and checked by script against the extraction table (48 of 48 rows matched). Comparative analysis drafted from it. *Done ahead of schedule on 13–14 Aug (NTCC diary); reported in this week's WPR.*	Master table (48 studies); comparative analysis draft; WPR 9 (dated 23 Sep)
10	21–27 Sep	Draft the challenges and gap analysis, the conclusion and the abstract.	Challenges and gap analysis, conclusion and abstract drafted; review text complete (about 9,850 words). The Semester 8 study appears as future work only, with no results. *Done ahead of schedule on 13–14 Aug (NTCC diary); reported in this week's WPR.*	Challenges, conclusion and abstract drafts; WPR 10 (dated 30 Sep)
11	28 Sep – 4 Oct	Assemble the full draft. Verify the references. Run consistency checks.	Full draft assembled as draft v1 (DOCX and 31-page PDF) with 73 cited references in IEEE style. Consistency checks run; all 74 reference checks confirmed. *Assembled ahead of schedule on 15–17 Aug, references checked on 10–12 Aug (NTCC diary); reported in this week's WPR. A journal-style draft v2 (19 pages) went to the department's plagiarism check on 15 Aug.*	Draft v1 (DOCX and PDF); citation map; WPR 11 (dated 7 Oct)
12	5–9 Oct	Guide review of the draft; corrections; restyling in the Amity format; submission pack.	Window closed on 9 Oct; no new deliverable. Submission pack reviewed (drafts v1 and v2, verified references, WPRs 1–11, Progress Reports 1 and 2, NTCC diary). Open at the close: the official report format, the plagiarism-check result, the team's own rewrite and the guide's review.	WPR 12 (dated 14 Oct), reported as incomplete
After the window (October 2026)		Prepare the final report in the Amity format.	Official report format (ETMN100 guidelines) obtained. Final report built on 20 Oct in the Amity "Project Plan" format: 100 pages, 15 chapters. Upload on Amizone due 26 Oct.	Final report (PDF, with DOCX source)

Note: this diary covers the literature review, which reports no experiments or results of its own. On 15 August 2026 the guide also asked the team to begin preparing the Semester 8 implementation code; that is a separate work track and is not recorded here.

Key dates

29 Jul 2026	Synopsis entered on Amizone.
27 Aug 2026	First progress review presentation (Progress Report 1 and the NTCC project diary were prepared for it).
1 Oct 2026	Progress Report 2 dated (progress up to the end of Week 10).
9 Oct 2026	Project window closed.
20 Oct 2026	Final report built in the Amity format.
26 Oct 2026	Final report upload on Amizone due.
2–6 Nov 2026	Evaluation.

Rohit Yadav Student Date: _____	**Vishal** Student Date: _____	**Hardik Mehlawat** Student Date: _____	**Dr. Richa Gupta** Faculty guide Date: _____